

Today : Finish §2.3

Def = The uncertainty of an observable  $A$  in the state  $|\psi\rangle$  is defined as

$$\Delta_\psi(A) = \sqrt{\langle \psi | (A - \langle A \rangle_\psi \mathbb{1})^2 | \psi \rangle}$$

$$= \sqrt{\langle (A - \langle A \rangle_\psi \mathbb{1})^2 \rangle_\psi}$$

$$\stackrel{(\ast)}{=} \sqrt{\langle A^2 \rangle_\psi - \langle A \rangle_\psi^2}$$

} Standard deviation of  $A$ .

•  $\Delta_\psi(A) = 0 \Rightarrow$  The value of the observable  $A$  in the state  $|\psi\rangle$  is sharp.

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Since  $(\Delta_\psi(A))^2 = \langle (A - \langle A \rangle_\psi \mathbb{1})^2 \rangle_\psi$  → scalar valued

$$= \langle A^2 - 2\langle A \rangle_\psi A + \langle A \rangle_\psi^2 \rangle_\psi$$

$$= \langle A^2 \rangle_\psi - \langle A \rangle_\psi^2$$


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$$\boxed{\Delta_\psi(A) = 0} \iff \langle A^2 \rangle_\psi = \langle A \rangle_\psi^2$$

Let  $|\lambda\rangle$  be an eigen-state with

$$A|\lambda\rangle = \lambda|\lambda\rangle \quad \Lambda: \text{set of eigen values.}$$

$$\Rightarrow \langle A^2 \rangle_\psi = \langle \psi | A^2 | \psi \rangle \quad \boxed{\mathbb{I} = \sum_{\lambda \in \Lambda} |\lambda\rangle \langle \lambda|}$$

$$= \sum_{\lambda_1, \lambda_2 \in \Lambda} \langle \psi | \lambda_1 \rangle \underbrace{\langle \lambda_1 | A^2 | \lambda_2 \rangle}_{\lambda_2^2 \langle \lambda_1 | \lambda_2 \rangle} \langle \lambda_2 | \psi \rangle$$

$$= \sum_{\lambda_1, \lambda_2 \in \Lambda} \langle \psi | \lambda_1 \rangle \lambda_2^2 \underbrace{\langle \lambda_1 | \lambda_2 \rangle}_{\delta_{\lambda_1 \lambda_2}} \langle \lambda_2 | \psi \rangle.$$

$\boxed{\{|\lambda\rangle\}: \text{ONB}}$

$$= \sum_{\lambda \in \Lambda} |\langle \psi | \lambda \rangle|^2 \lambda^2.$$

$$\langle A \rangle_\psi = \langle \psi | A | \psi \rangle = \sum_{\lambda \in \Lambda} |\langle \psi | \lambda \rangle|^2 \lambda.$$

$$\boxed{\langle A^2 \rangle_\psi = \langle A \rangle_\psi^2} \Rightarrow \sum_{\lambda \in \Lambda} |\langle \psi | \lambda \rangle|^2 \lambda^2 = \left( \sum_{\lambda \in \Lambda} |\langle \psi | \lambda \rangle|^2 \lambda \right)^2$$

$P_\psi(\lambda) = |\langle \psi | \lambda \rangle|^2$  (Probability), Hölder Ineq.

$$\left( \sum_{\lambda} P_\psi(\lambda) \lambda^2 \right) \left( \sum_{\lambda} P_\psi(\lambda) \right) \geq \left( \sum_{\lambda} P_\psi(\lambda) |\lambda| \right)^2$$

$\downarrow$   $\downarrow$   $\left( \sum_{\lambda} P_\psi(\lambda) \lambda \right)^2 = \langle A \rangle_\psi^2$   
 $\langle A^2 \rangle_\psi$  Triangle Ineq. condition of.

$\Rightarrow$  Now, we want to find equality holds.

Hölder - Inequality

$$\frac{P_{\psi}(\lambda) \lambda^2}{P_{\psi}(\lambda)} = \text{const.} \Rightarrow \lambda^2 \text{ should be constant.}$$

Triangle Inequality,

Signs of  $\lambda \in \Lambda$  should be same.

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$$\Rightarrow \exists! \lambda \in \Lambda \text{ s.t. } P_{\psi}(\lambda) \neq 0.$$

$$\Rightarrow \boxed{\psi \text{ is an eigen state of } A.}$$

$$\rightarrow \text{Result of } \boxed{\text{Proposition 2.17.}}$$

Def Two observables  $A, B$  are called

$$\begin{cases} \text{compatible if } [A, B] = 0. \\ \text{Incompatible if } [A, B] \neq 0. \end{cases}$$

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Prop. (Uncertainty Principle)

For any observables  $A, B \in \mathcal{B}_{\text{sa}}(\mathcal{H})$  and state  $|\psi\rangle \in \mathcal{H}$  the following uncertainty relation holds.

$$\Delta_{\psi}(A) \cdot \Delta_{\psi}(B) \geq \left| \left\langle \frac{1}{2i} [A, B] \right\rangle_{\psi} \right|.$$

Idea of proof  $\Rightarrow$  Just Hölder inequality.

e.g.  $A = \hat{x}$ ,  $B = \hat{p}$ .

Since  $[\hat{x}, \hat{p}] = i\hbar$ , we have

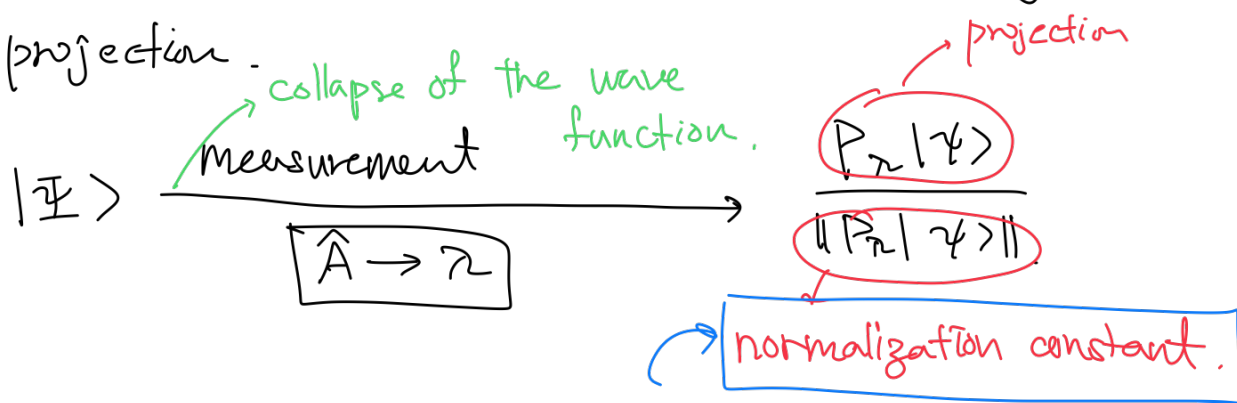
$$\Delta_{\psi}(\hat{x}) \cdot \Delta_{\psi}(\hat{p}) \geq \left| \left\langle \frac{1}{2i} \cdot i\hbar \right\rangle_{\psi} \right|^{\frac{1}{2}} = \frac{\hbar}{2}$$

position                  momentum,

$\Rightarrow$  We cannot measure position and momentum simultaneously.

Postulate 3. (Projection Postulate)

If a measurement of the observable  $A$  on a quantum mechanical system in the pure state  $|\psi\rangle \in \mathcal{H}$  yields the eigenvalue  $\lambda$ , then the measurement has affected the following state projection.



(Every quantum state satisfies  $\|\psi\|_2 = 1$ )

Let  $\{\lambda \in \Lambda\}$  s.t.  $\hat{A}|\lambda\rangle = \lambda|\lambda\rangle$   
 $\downarrow$   
 $\boxed{\text{ONB}}$

If we set

complex number  
 $|c|=1$   $\boxed{c = e^{i\phi}}$

$$|\Psi\rangle = \sum_{\lambda \in \Lambda} c_{\lambda} |\lambda\rangle \xrightarrow[\hat{A} \rightarrow \lambda_0]{\text{measurement}} |\lambda_0\rangle$$

$\boxed{c_{\lambda_0} \neq 0}$

\* Time evolution.

$U(t_2, t_1)$  is a time-evolution operator:

$$|\psi(t_1)\rangle \xrightarrow[\text{time-evolution.}]{\text{time-evolution.}} |\psi(t_2)\rangle = U(t_2, t_1) |\psi(t_1)\rangle$$

(No measurement)

Prop. ①  $U(t_2, t_1)$  should be unitary.  $\forall t_1 < t_2$ .

$$\textcircled{2} \begin{cases} i \frac{d}{dt} U(t, t_0) = H(A) U(t, t_0) & (H(A) \in \mathcal{B}_{sa}(H)) \\ U(t_0, t_0) = \mathbb{I} \end{cases} \rightarrow \dot{U}(t, t_0) = -i H(A) U(t, t_0)$$

(pf). ① Since  $\| |\psi(t_1)\rangle \| = \| |\psi(t_0)\rangle \| = 1$ ,  $U(t_2, t_1)$  should be unitary.  $H(A) = H^*(A)$

$$\begin{aligned} \textcircled{2} \frac{d}{dt} (U^*(t, t_0) U(t, t_0)) &= \dot{U}^*(t, t_0) U(t, t_0) \\ &= (-i U^*(t, t_0) H^*(A)) U(t, t_0) - U^*(t, t_0) (i H(A) U(t, t_0)) \\ &= 0. \end{aligned}$$

$$\underline{U^\dagger(t, t_0) U(t, t_0)} = U^\dagger(t_0, t_0) U(t_0, t_0) \\ = \mathbb{1} \cdot \mathbb{1} = \mathbb{1}.$$

$\Rightarrow U(t, t_0)$  is an unitary operator.

□

Postulate 4 (Time-evolution)

$|\psi(t_0)\rangle$ : state at time  $t_0$

$\xrightarrow{\text{No measurement}} |\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle$   
: state at time  $t$ .

Recall that, Schrödinger Eq

$$i\hbar \partial_t |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

$$\Rightarrow i\hbar \partial_t (U(t, t_0) |\psi(t_0)\rangle) = H(t) U(t, t_0) |\psi(t_0)\rangle$$

$\hookrightarrow$  does not depend on " $t_0$ "

$$\Rightarrow (i\hbar \partial_t U(t, t_0)) |\psi(t_0)\rangle = (H(t) U(t, t_0)) |\psi(t_0)\rangle$$

$$\Rightarrow i\hbar \partial_t U(t, t_0) = H(t) U(t, t_0).$$

## Spin

• Important concept in Q.M.

• Each particle (electron, proton, photon, neutron, ...) has a different spin structure.

• In Quantum Computation, we use "electron"

So in this book, we'll study about spin of electron.

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In classical computer  $\Rightarrow$   $\begin{matrix} 0 \text{ or } 1. \\ \begin{matrix} \boxed{0} \\ \boxed{1} \end{matrix} \text{ or } \begin{matrix} \boxed{S} \\ \boxed{N} \end{matrix} \end{matrix} \rightarrow \boxed{\text{bit}}$

In Quantum computer  $\Rightarrow \boxed{\text{Qubit}} \Rightarrow \boxed{\text{Express a state from electron.}}$

~~Q~~ Spin is abstract concept, but physical meaning of spin is similar to angular momentum of quantum particles.

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We'll study angular momentum in Q.M.

In classical mechanics,

even perm  
of (1,2,3)

$$\vec{L} = \vec{r} \times \vec{p} \quad \text{Cross product.}$$

Levi-Civita Symbol.

$$L_i = (\vec{r} \times \vec{p})_i = \epsilon_{ijk} r_j p_k$$

$$\begin{cases} +1 & \text{if } (i,j,k) \\ -1 & \text{if } (i,j,k) \end{cases}$$

odd perm.

$$(1=x, 2=y, 3=z)$$

$$\epsilon_{113} = 0$$

if  $\exists$  repeated index

$L_3$ : z-component of angular momentum vector  $\vec{L}$

$$\begin{cases} L_x = y p_z - z p_y \\ L_y = z p_x - x p_z \\ L_z = x p_y - y p_x \end{cases} \Rightarrow \begin{cases} \hat{L}_x = \hat{y} \hat{p}_z - \hat{z} \hat{p}_y \\ \hat{L}_y = \hat{z} \hat{p}_x - \hat{x} \hat{p}_z \\ \hat{L}_z = \hat{x} \hat{p}_y - \hat{y} \hat{p}_x \end{cases}$$

$$\hat{L}_i = \epsilon_{ijk} \hat{r}_j \hat{p}_k$$

$$\begin{aligned} L_1 &= L_x, L_2 = L_y, L_3 = L_z \\ x_1 &= x, x_2 = y, x_3 = z. \\ p_1 &= p_x, p_2 = p_y, p_3 = p_z. \end{aligned}$$

$$\begin{aligned} [x_i, x_j] &= 0 \\ [p_i, p_j] &= 0 \\ [x_i, p_j] &= i\hbar \delta_{ij} \\ \text{if } i=j &\Rightarrow [x, p_x] = [y, p_y] = [z, p_z] = i\hbar \\ \text{if } i \neq j &\Rightarrow [x, p_y] = 0 \end{aligned}$$

Einstein Notation.

tensor contraction

$$\epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl} \quad \text{Tensor Identity.}$$

rank-4 tensor.

$\epsilon_{ijk}$ : rank-3 tensor.

$$j, l, k, m \in \{1, 2, 3\}$$

$$3 + 3 - 2 = 4$$

$$\Rightarrow 3^4 = 81$$



$$[\hat{L}_i, \hat{L}_j] = [\underbrace{\epsilon_{ilm}} \hat{x}_l \hat{p}_m, \underbrace{\epsilon_{jkn}} \hat{x}_k \hat{p}_n]$$

$$= \epsilon_{ilm} \epsilon_{jkn} [\hat{x}_l \hat{p}_m, \hat{x}_k \hat{p}_n]$$

$$= ( \quad ) (\hat{x}_l \hat{p}_m \hat{x}_k \hat{p}_n - \hat{x}_k \hat{p}_n \hat{x}_l \hat{p}_m)$$

$$\begin{aligned} & \hat{x}_l [\hat{p}_m, \hat{x}_k] \hat{p}_n + \hat{x}_l \hat{x}_k \hat{p}_m \hat{p}_n \\ & - \hat{x}_k [\hat{p}_n, \hat{x}_l] \hat{p}_m - \hat{x}_k \hat{x}_l \hat{p}_n \hat{p}_m \end{aligned}$$

$\xrightarrow{-i\hbar \delta_{mk}} \quad \xrightarrow{-i\hbar \delta_{nl}}$

$= 0 \quad (\because [\hat{x}_l, \hat{x}_k] = 0, [\hat{p}_m, \hat{p}_n] = 0)$

$$= \epsilon_{ilm} \epsilon_{jkn} (i\hbar \delta_{nl} \hat{x}_k \hat{p}_m - i\hbar \delta_{mk} \hat{x}_l \hat{p}_n)$$

$$= i\hbar \left( \underbrace{\epsilon_{ilm} \epsilon_{jkl}}_{\downarrow} \hat{x}_k \hat{p}_m - \underbrace{\epsilon_{ilk} \epsilon_{jkn}}_{\downarrow} \hat{x}_l \hat{p}_n \right)$$

$$= i\hbar \left( \underbrace{\epsilon_{lmj} \epsilon_{ljk}}_{\downarrow} \hat{x}_k \hat{p}_m + \underbrace{\epsilon_{kij} \epsilon_{kin}}_{\downarrow} \hat{x}_l \hat{p}_n \right)$$

$$\boxed{\delta_{mj} \delta_{lk} - \delta_{mk} \delta_{lj} \quad \delta_{ij} \delta_{kn} - \delta_{in} \delta_{kj}}$$

= ...

$$= i\hbar (\hat{x}_i \hat{p}_j - \hat{x}_j \hat{p}_i) = i\hbar \epsilon_{ijk} \hat{L}_k$$

$$\therefore [\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k$$

Relation between Angular momentum operator.

Now, come back to "Spin".

**Def** • The matrices  $\sigma_j \in \text{Mat}_{2 \times 2}(\mathbb{C})$  indexed by either  $\hat{j} \in \{1, 2, 3\}$  or  $\hat{j} \in \{x, y, z\}$  and defined as

$$\sigma_x = \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_z = \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\Rightarrow$  Pauli-matrices,  $(\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix})$ ,

$$\{\sigma_1\} = \{\sigma_1, \sigma_2, \sigma_3\}.$$

$$\{\sigma_2\} = \{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$$

•  $\hat{S}_j = \frac{1}{2} \sigma_j$  spin operator for electron.

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From the simple calculation, we have

$$[\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k.$$

SI

$$[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k$$

$$\hat{L}_i = \frac{2}{\hbar} \sigma_i$$

$\hookrightarrow$  We use Pauli-Matrices to express spin operator.

$$\begin{aligned} |\uparrow_z\rangle &= |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |\downarrow_z\rangle &= |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned} \Rightarrow \begin{cases} \hat{S}_z |\uparrow_z\rangle = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2} |\uparrow_z\rangle \\ \hat{S}_z |\downarrow_z\rangle = -\frac{1}{2} |\downarrow_z\rangle. \end{cases}$$

$|\uparrow_z\rangle, |\downarrow_z\rangle \Rightarrow$  eigen state of  $\hat{S}_z$ .  $\frac{1}{2}, -\frac{1}{2}$  resp.

Remark.

Since  $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k \neq 0$  for  $i \neq j$ ,

$\sigma_i, \sigma_j$  are Incompatible.

$\Leftrightarrow \hat{S}_i, \hat{S}_j$  are Incompatible  $\forall i \neq j$ .

## 2.3.2 Mixed State.

Overview • pure state:  $|\psi\rangle \in \mathcal{H}$  with  $\|\psi\|=1$ .

• mixed state:  $\rho \in \mathcal{B}_{S, \infty}(\mathcal{H})$  with  $\begin{pmatrix} \text{tr}(\rho) = 1 \\ \rho \geq 0 \\ \vdots \end{pmatrix}$

density operator.

$$\rho = \sum_{\alpha \in I} p_{\alpha} |\psi_{\alpha}\rangle \langle \psi_{\alpha}|$$

with  $\sum_{\alpha} p_{\alpha} = 1, \{|\psi_{\alpha}\rangle\} : \text{O.O.B.}$

Remark. Density operator for pure state  $|\psi\rangle$

$$\Rightarrow \boxed{\rho = |\psi\rangle \langle \psi|}$$

## Postulate 5 (Mixed State)

$\rho$ : density operator satisfies following properties.

- ①  $\rho$ : self-adjoint  $\rho^* = \rho$ .
  - ② positive  $\boxed{\rho \geq 0}$  ( $\langle \psi | \rho | \psi \rangle \geq 0, \forall |\psi\rangle \in H$ )
  - ③  $\rho$  has trace 1.  $\text{tr}(\rho) = 1$ .
- $\Rightarrow D(H)$ : set of density operators defined on  $H$ .

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Remark. convex combination.

- ①  $\rho_1, \rho_2 \in D(H) \Rightarrow \boxed{\lambda \rho_1 + (1-\lambda) \rho_2} \in D(H) (0 \leq \lambda \leq 1)$
- ②  $\rho \in D(H) \Rightarrow U \rho U^* \in D(H), \forall$  unitary  $U$ .

## Postulate 6.

(just generalization of  $\boxed{\text{Postulate 1-4}}$ ) ↗ (pure state)

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$\boxed{\text{Thm 2.24}} \quad \rho \in D(H)$

- ①  $\exists P_j$ , ONB  $\{|\psi_j\rangle\}$  s.t

$$\rho = \sum_j P_j |\psi_j\rangle \langle \psi_j|.$$

- 
- ②  $0 \leq \rho^2 \leq \rho$ ,

- ③  $\|\rho\|_{\text{op}} \leq 1$ .

(pf) ① omit. (infinite dim - linear algebra).

$$\textcircled{2} \quad \rho = \sum_j p_j |\psi_j\rangle \langle \psi_j|$$

$$\begin{aligned} 0 \leq \rho^2 &= \sum_{j,k} (p_j |\psi_j\rangle \langle \psi_j|) (p_k |\psi_k\rangle \langle \psi_k|) \\ &= \sum_{j,k} p_j p_k |\psi_j\rangle \underbrace{\langle \psi_j | \psi_k \rangle}_{\delta_{j,k}} \langle \psi_k| \\ &= \sum_j (p_j^2) |\psi_j\rangle \langle \psi_j|. \end{aligned}$$

$$\rightarrow \leq \sum_j (p_j) |\psi_j\rangle \langle \psi_j| = \rho.$$

$$\textcircled{3} \quad \|\rho\|_{\infty}^2 = \sup_{\|\psi\|=1} |\rho|\psi\rangle|$$

Mistake  
in note.

$$\left| \sum_j p_j |\psi_j\rangle \langle \psi_j| \psi \right|^2 \leq \sum p_j$$

Prop. 2.25/2.26.

$\rho \in D(H)$  is a pure state  $\iff \rho = \rho^2$ .

$(\Rightarrow)$  Easy.

$\|\psi\|=1$  for pure state

$$\begin{aligned} \text{Let } \rho = |\psi\rangle \langle \psi| &\Rightarrow \rho^2 = |\psi\rangle \langle \psi | \psi \rangle \langle \psi| \\ &= |\psi\rangle \langle \psi| = \rho. \end{aligned}$$

$$(\Leftarrow) \rho^2 = \sum_j p_j^2 |\psi_j\rangle\langle\psi_j|$$

$$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$$

If  $\rho^2 = \rho \Rightarrow$  Matrix representation of  $\rho^2, \rho$  should be same.

$$\Rightarrow \boxed{p_j^2 = p_j} \quad \bigg/ \quad \sum_j p_j = 1$$

$\Rightarrow$  only one  $i \in \Lambda$  s.t.  $p_i = 1$ .

$\forall j \neq i$  s.t.  $p_j = 0$

$$\Rightarrow \rho = p_i |\psi_i\rangle\langle\psi_i| = |\psi_i\rangle\langle\psi_i| \Rightarrow \boxed{\text{pure state}}$$

Prop 2.29. Let  $H$  be a finite-dimensional Hilbert space and  $\rho \in \mathcal{D}(H)$  be a density operator with diagonal form

$$\rho = \sum_{j=1}^n p_j |\psi_j\rangle\langle\psi_j| \quad \text{where } p_j \in (0, 1] \quad \forall j = \{1, \dots, n\}$$

with  $n \leq \dim(H)$  are the nonzero eigenvalues of  $\rho$ .

$\{|\psi_j\rangle\}_{j=1}^n$  are its eigenvectors.

Moreover,  $\forall i \in \{1, \dots, m\}$  with  $m \leq \dim(H)$ ,  $q_i \in (0, 1]$

$\sum_{i=1}^m q_i = 1$ ,  $|\varphi_i\rangle \in H$ ,  $\|\varphi_i\| = 1$ . Then

$$\boxed{\rho = \sum_{i=1}^m q_i |\varphi_i\rangle\langle\varphi_i|} \iff \left( \begin{array}{l} m \geq n, \exists U \in \mathcal{U}(m) \\ \sqrt{q_i} |\varphi_i\rangle = \sum_{j=1}^n U_{ij} \sqrt{p_j} |\psi_j\rangle \\ \forall i = 1 \dots m \end{array} \right)$$

(PL) Omit. (Linear Algebra)

